SESSION 2 · WED 21 OCTOBER 2026

Strategic Asset Allocation

From beliefs to a robust strategic portfolio

Portfolio Management (28E34600), Autumn 2026
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Reading: study notes, Session 2; lecture notes Ch. 6–9

Agenda

1. What exactly is strategic asset allocation?
2. What does a real strategic portfolio look like?
3. What happens if we just optimize?
4. How do we make the strategic portfolio robust?
5. What must your Case 1 strategic portfolio contain?
6. Wrap-up and next steps

Part 1

1 **What exactly is strategic asset allocation?**

2 What does a real strategic portfolio look like?

3 What happens if we just optimize?

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6 Wrap-up and next steps

From Session 1: theory works only with disciplined inputs

What we learned

60/40 is about 90% stocks in risk terms. Expected returns are imprecise, and an optimizer takes their errors literally. Optimal weights are random variables.

Today

How a long-term investor turns beliefs, risk and policy into a strategic portfolio that is *robust*: sensible even if the inputs are somewhat wrong.

The question for Case 1: what should a Finnish pension fund own for the long run, and why?

Three decisions, three horizons

	Strategic (SAA)	Tactical (TAA)	Implementation
Question	What do we own in the long run?	Do we deviate from it now?	How do we get the exposure?
Horizon	5–10 years and more	months to a few years	days to months
Inputs	CMAs, risk, risk tolerance, policy	views and their confidence	instruments, costs
Output	policy weights	tilts of a few points	funds, futures, trades
In Case 1	Task 1, Sessions 2–5	Task 2, Sessions 4–5	in the report

Keep the three apart. Most of the variation in a fund's returns comes from the strategic decision (Brinson et al., 1986).

The strategic problem in one line



$$w_{\mathrm{SAA}} = \arg\max_{w \in \mathcal{W}} \; w^{\top}\mu - \frac{\gamma}{2} w^{\top}\Sigma w$$

Each of the four inputs is a choice that must be stated and defended. The optimizer only combines them.

An SAA is beliefs plus risk plus policy. Write down the beliefs first.

Investment beliefs come first

What a belief is

Which sources of return should be rewarded over the long run, and why? For example: equity investors earn a premium for bearing recession risk; illiquidity is rewarded only for those who can hold.

Beliefs are forward-looking

History informs them, but a historical average is not a belief about the future (Session 1).

Start from an anchor

The market portfolio (Session 5) or the current portfolio. Beliefs explain why the fund should move away from it, and by how much.

Evidence and dates

Each belief needs a reason, evidence and a date, and a statement of what would change your mind.

Poll 1

The Case 1 pension fund has 50% of its money in equities and real assets. Roughly what share of its *risk* comes from them?

A about 50%

B about 65%

C about 75%

D about 90%



Scan to vote

Part 2

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Goldman Sachs Balanced Strategy: a real multi-asset fund

The fund

US mutual fund managed by Goldman Sachs Asset Management; objective: current income and long-term growth. About USD 609 million at 31 December 2025. The Institutional class (GIPIX) supplies the return series.

The strategic starting point

Component	Normal	Minimum	Maximum
Equity	35%	5%	60%
Fixed income	50%	25%	80%
Dynamic	15%	0%	20%

The **normal allocation** is the SAA; the **ranges** leave room for tactical deviation.

Goldman Sachs Balanced Strategy Portfolio, summary prospectus 2025; lecture notes Ch. 6.

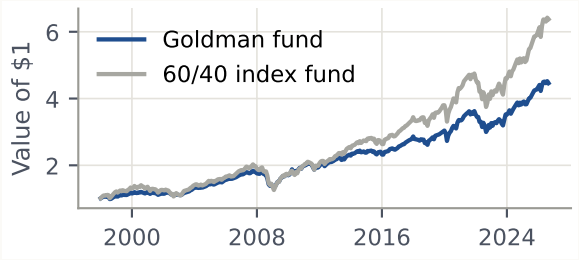
The portfolio on 31 December 2025 is not the strategic portfolio

Exposure	Weight	Main components
Equity	45.3%	US 27.9%, developed ex-US 11.3%, EM 4.4%, real assets 1.6%
Fixed income	56.1%	investment grade 52.0%, high yield 2.4%, EM debt 1.8%
Multi-strategy	2.7%	managed futures 1.0%, tactical tilts 1.7%
Cash and other	−4.0%	offsets derivative exposure above 100%

- Equity at 45.3% is above the 35% normal weight but inside the 5–60% range: a tactical position
- Exposures add up to 104%: futures create exposure without the same cash investment
- The *allocation* fixes the exposure; *implementation* chooses the instrument

Goldman Sachs Asset Management, holdings report Q4 2025; lecture notes Ch. 6.

Did Goldman add value?



1998–2026: Goldman 5.3% a year (volatility 7.6%); 60/40 fund 6.6% (10.0%).

−1.5%

a year relative to a passive 60/40 fund

Of that gap:

less market risk (beta 0.73)	−1.3%
alpha, risk-adjusted	−0.1%

Alpha not different from zero ($t = -0.4$).

Lower return came from holding less risk, a strategic choice, not from a lack of skill.

GIPIX and VSMGX, monthly, net of fees. Official benchmark: 40% MSCI ACWI + 60% Bloomberg Global Aggregate; the 60/40 fund is a stand-in.

Return = strategy + tactics + selection

Active return

$r_{\text{fund}} - r_{\text{bench}}$

Simple, but it mixes skill with differences in risk.

Alpha

$r_{\text{fund}} - r_f = \alpha + \beta\,(r_{\text{bench}} - r_f) + \varepsilon$

What is left after adjusting for the exposure β. Only α measures skill.

Where a fund's return comes from

1. **Strategy:** the benchmark or normal allocation. Usually most of the return and risk
2. **Tactics:** deviations such as Goldman's 45% equity against 35% normal
3. **Selection and costs:** the funds and securities chosen, minus fees

Session 5 sizes tactical tilts; Case 2 judges your fund by its alpha and information ratio.

Judge a manager by alpha, not by raw return against a benchmark.

Part 3

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Nine asset classes: beliefs and risk

The same nine classes as Case 1; expected returns from Goldman's public long-run forecasts (nominal, USD)

Asset class	CMA μ	Volatility	Current
US equity	6.5%	15.2%	25%
Developed ex-US equity	8.0%	16.8%	15%
Emerging-markets equity	10.9%	20.5%	5%
Global real assets	8.7%	20.5%	5%
Investment-grade bonds	4.3%	4.2%	30%
High-yield bonds	5.9%	7.6%	5%
Emerging-markets debt	3.6%	10.1%	5%
Managed futures	3.1%	10.4%	5%
Cash	3.15%	0.6%	5%

Correlations that matter

- US and developed equity: 0.86
- US and EM equity: 0.74
- US equity and IG bonds: 0.12
- US equity and managed futures: −0.15

Current portfolio: expected return 5.9%, volatility 8.6%.

CMAs: Goldman Sachs Asset Management long-run forecasts mapped to the nine classes (lecture notes Ch. 8). Risk: Case 1 monthly returns 2000–2025, USD, annualized. Your Case 1 CMAs will differ.

Poll 2

With these inputs, $\gamma = 4$ and only "fully invested, no short positions", how much US equity does the optimizer choose?

A 0%

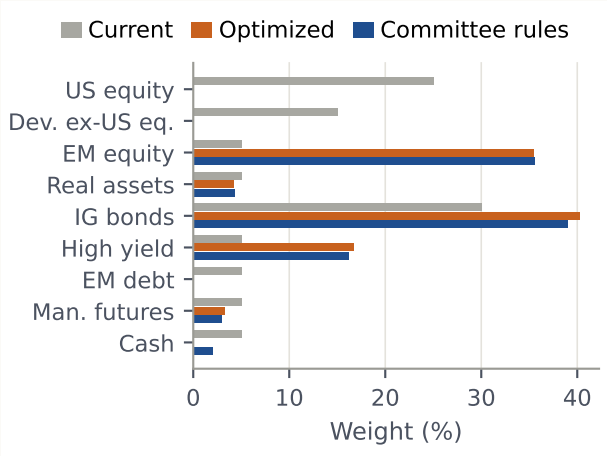
B about 10%

C about 25%, as now

D about 40%

Scan to vote

Poll 2 revealed: no US equity at all



Answer A. The optimizer holds three asset classes in size: EM equity 36%, IG bonds 40%, high yield 17%, plus small positions in real assets and managed futures.

Expected return 7.1%, volatility 9.1%.

Adding the Committee's rules (cash at least 2%, nothing above 40%) changes almost nothing: the rules hardly bind at $\gamma = 4$.

Corner solutions are not calculation errors: they follow from the inputs.

Why the optimizer picks what it picks

Equities compete with each other

US, developed and EM equity are highly correlated. The optimizer keeps the one with the best return per unit of shared risk: EM, with the highest CMA (10.9%).

Small differences, big moves

A 1.5-point lower CMA (US 6.5% vs developed 8.0%) is enough to remove US equity completely. That gap is well inside the uncertainty of any CMA.

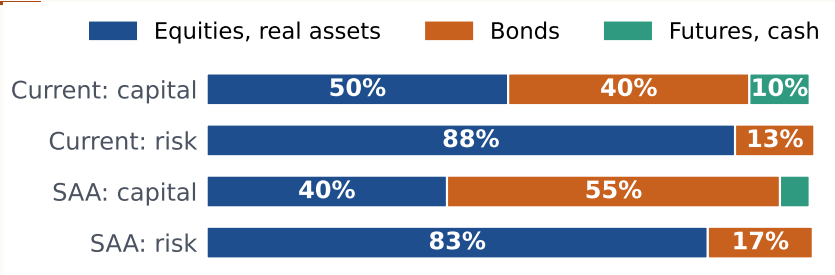
Bonds diversify

IG bonds have low volatility and low correlation with equities, so they take most of the remaining weight.

What is left out

Estimation error, liquidity, drawdowns, currency and governance do not enter unless we add them.

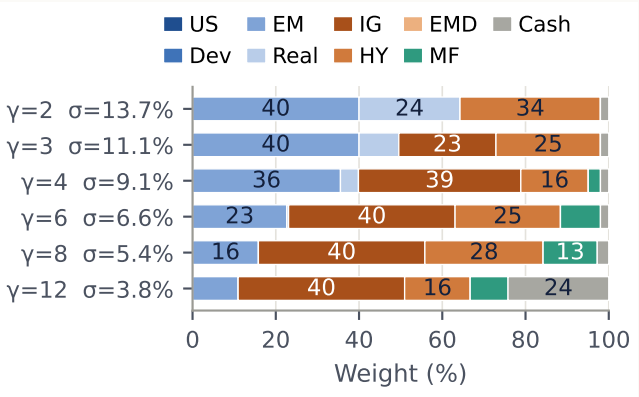
Poll 1 revealed: risk shares, not weights



- **Answer D.** In the current portfolio, 50% of the money in equities and real assets carries 88% of the risk
- The optimized SAA is no better: 40% of the money, 83% of the risk, almost all in EM equity
- Same tool as Session 1: $RC_i = w_i(\boldsymbol{\Sigma w})_i/\sigma_p$. Report risk contributions, not only weights

Risk contributions with the Case 1 covariance matrix. Managed futures have a slightly negative risk contribution.

Risk aversion changes the strategic portfolio



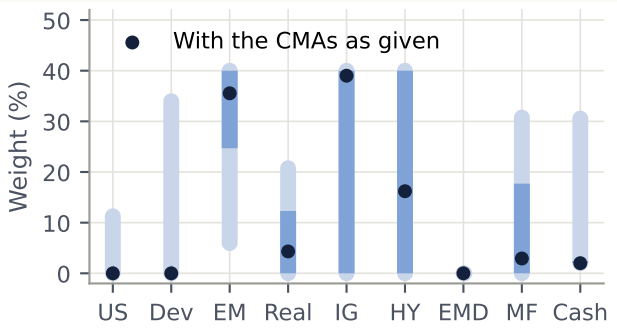
Committee rules imposed: fully invested, no short positions, cash at least 2%, no class above 40%.

- γ from 2 to 12: volatility from 13.7% to 3.8%
- US and developed equity stay at zero for every γ
- **Two rules:** the γ that gives the risk the fund already accepts (8.6% $\Rightarrow \gamma \approx 4.25$), or the market's $\gamma = 2.5$ (Session 5)
- Then show the neighbouring values

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If the CMAs are 1 point off, the portfolio changes completely



Add random errors with a standard deviation of only 1 percentage point to each CMA and re-optimize, 1,000 times.

- IG and high yield range from 0% to 40%
- Each is *zero* in about a third of the cases
- Managed futures: 0% to 31%

Bars: 5th–95th percentile; darker part: middle half.

The strategic portfolio is an estimate too. A robust SAA should not depend on CMA differences we cannot measure.

Five ways to make the SAA robust

Remedy	What it does
1. Better beliefs	forward-looking CMAs from yields, growth and valuation, stated as a range, not a point (Sessions 3 and 4)
2. Shrink toward an anchor	pull noisy CMAs toward the returns implied by an anchor (here the current portfolio); Black–Litterman does it formally from market weights (Session 5). Also for $\boldsymbol{\Sigma}$ (Ledoit and Wolf, 2004)
3. Resample	average the optimal weights over many plausible CMAs (Michaud, 1998)
4. Constraints with a reason	limits for liquidity, governance or concentration also damp the optimizer's reaction to errors
5. Simple benchmarks	compare with the current portfolio, $1/N$ and equal risk contributions; out of sample, optimized portfolios rarely beat $1/N$ (DeMiguel, Garlappi and Uppal, 2009)

Comparing the candidates

Portfolio	Return	Volatility	Largest	Classes >5%	Equity risk
Current (reference)	5.9%	8.6%	30%	3	88%
Mean–variance, Committee rules	7.0%	9.1%	39%	3	83%
Shrunk CMAs	6.9%	8.7%	34%	4	78%
Resampled average	6.9%	9.1%	30%	6	84%
Equal risk contributions	4.9%	5.2%	33%	5	50%
$1/N$	6.0%	8.7%	11%	9	81%

- Robust methods give up a few tenths of a percent of *estimated* return for a much more diversified portfolio
- The estimated return is the least reliable number in the table

Expected returns with the Goldman-based CMAs; $\gamma = 4$; Committee rules in all optimized portfolios. Shrunk CMAs: halfway to reference-implied returns. Largest: largest weight. Equity risk: risk contribution of equities and real assets.

Constraints: investment policy written as algebra

Good constraints

Have an investment, risk, liquidity or governance reason: a cash floor for pension payments, a cap so that no single class dominates, a limit on illiquid assets.

Does a constraint bind?

If the unconstrained solution already satisfies it, it costs nothing. If it binds, it lowers the estimated objective: report which limits bind and why the rationale is worth the cost.

Cosmetic constraints

Added only to make an optimized portfolio look diversified. They hide bad inputs instead of fixing them. Case 1 does not allow them.

Separate from beliefs

Constraints are policy. Do not use them to correct CMAs you do not believe: change the CMAs.

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Case 1: the Committee's rules

The rules

$\mathbf{1}^\top \boldsymbol{w} = 1$ fully invested
$w_i \geq 0$ no short positions
$w_{\text{Cash}} \geq 0.02$ liquidity for pensions
$w_i \leq 0.40$ no class dominates

- EUR 10 billion Finnish pension fund, euro liabilities, long horizon
- The current portfolio is a starting point, not a target
- Extra constraints only with a stated reason
- Your CMAs: expected excess returns over cash at the case date, 30.9.2026, from yields, growth and valuation (Session 3), not Goldman's

What the Committee needs to see in Task 1

Item	What to show	Today's example
Beliefs and CMAs	nine CMAs, their basis and date	Goldman-based, USD
Risk model	volatilities, key correlations	sample 2000–2025
Risk tolerance	γ and why	4.25 (today's 8.6% risk)
Portfolio	proposed vs current weights	7.0%, 9.1% vs 5.9%, 8.6%
Risk contributions	where the risk comes from	83% equities
Robustness	sensitivity to γ and CMAs	1-point CMA errors
Rationale	a story for every large or zero weight	why no US equity?

An optimized portfolio is not a recommendation until its drivers and risks are explained.

Case 1 timeline

Date	Session	For Case 1
Wed 21.10	SAA (today)	beliefs, risk model, strategic portfolio set-up
Tue 27.10	Expected returns I, with Antti Ilmanen (AQR)	CMAs; study notes 3
Wed 28.10	Expected returns II	views for Task 2
Tue 3.11	Black–Litterman: SAA and TAA	Task 2
Wed 4.11	Online clinic for Case 1 (Mikko Kauppila)	your questions; Case 1 exercise
Fri 13.11, 16:00		**report due**
Week of 16.11		presentations

Build the SAA pipeline now; add CMAs after Session 3 and the tactical step after Session 5.

Part 6

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Takeaways

SAA is the long-run decision

Beliefs, risk, risk tolerance and policy. Keep it apart from tactical views and implementation.

Raw optimization concentrates

No US equity, 36% EM; a 1-point CMA error can move a weight from 0% to 40%.

Strategy drives returns

Goldman lagged a 60/40 fund by 1.5 points a year, almost all from holding less risk; its alpha was about zero.

Robustness by design

Better CMAs, shrinkage, resampling, reasoned constraints and simple benchmarks. Report risk contributions.

Before next week

- **Read** the Session 2 study notes; lecture notes Ch. 6–9 for the details
- **Case 1:** write down your group's investment beliefs; load the monthly data and estimate the covariance matrix; set up the optimizer with the Committee's rules
- **Prepare** for Antti Ilmanen's session: read AQR (2026), *Capital Market Assumptions*, study notes 3 and lecture notes Ch. 15–21

Next: Session 3, Tue 27.10 – Expected returns I, with Antti Ilmanen (AQR).

References (backup)

- AQR (2026). *Capital Market Assumptions for Major Asset Classes*. Alternative Thinking.
- Brinson, G., L. R. Hood and G. Beebower (1986). Determinants of portfolio performance. *Financial Analysts Journal* 42(4).
- DeMiguel, V., L. Garlappi and R. Uppal (2009). Optimal versus naive diversification. *Review of Financial Studies* 22(5).
- Goldman Sachs Asset Management (2025). Balanced Strategy Portfolio: summary prospectus and holdings, Q4 2025.
- Ledoit, O., and M. Wolf (2004). Honey, I shrunk the sample covariance matrix. *Journal of Portfolio Management* 30(4).
- Michaud, R. (1998). *Efficient Asset Management*. Harvard Business School Press.